Mutation → Behavior Class Map v1.0

> **Scope:** All 21 genes from the Stroma World gene panel. **Format:** Each mutation's effect on six behavioral axes, described qualitatively. **No numeric multipliers.** Effects are directional: ↑ (increased), ↓ (decreased), — (negligible/no direct effect), or ◆ (context-dependent, see caveat). **Mutation type** reflects what actually happens in PDAC, not all possible mutations.

Effect Scale

Symbol	Meaning
↑↑↑	Strong increase — dominant driver of this axis
↑↑	Moderate increase — meaningful but not sole driver
↑	Mild increase — detectable, secondary
—	Negligible or no direct effect on this axis
↓	Mild decrease
↓↓	Moderate decrease
↓↓↓	Strong decrease
◆	Context-dependent — see caveat

Confidence Levels

Tag	Meaning
HIGH	Established in PDAC models and human tissue; multiple concordant sources
MED	Demonstrated in pancreatic cancer contexts but with some inconsistency or limited mechanistic clarity
LOW	Inferred from pan-cancer data or pathway logic; not yet rigorously validated in PDAC specifically

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Axis Definitions (for this map)

1. **Growth Drive (GD):** Rate of proliferative cycling under favorable conditions.
2. **Death Resistance (DR):** Ability to evade apoptosis, anoikis, and therapy-induced death.
3. **Cell Cycle Braking (CCB):** Capacity to undergo growth arrest in response to crowding, DNA damage, or anti-growth signals. *Higher = more responsive to brakes. Lower = brakes are broken.*
4. **Motility / Invasion (MI):** Migration speed, ECM remodeling, mesenchymal phenotype.
5. **Pro-Stroma Signaling (PSS):** Tendency to activate stellate cells, stimulate ECM deposition, build the barrier.
6. **Immune-Evasion Tendency (IET):** Ability to suppress or escape immune surveillance. *Included for completeness even though immune cells are out of scope for Cycle 1.*

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1 — GROWTH DRIVE AXIS

KRAS (gain-of-function, G12D/G12V/G12R)

GD	DR	CCB	MI	PSS	IET
↑↑↑	↑↑	↓↓	↑	↑↑↑	↑↑

- **Confidence:** HIGH across all axes.
- **Caveat:** KRAS is the root node. It doesn't just drive growth — it activates RAF/MEK/ERK (growth), PI3K/AKT (survival), and RalGEF pathways (invasion, secretion). The pro-stroma effect is indirect but powerful: KRAS drives secretion of SHH, TGF-β, and IL-6 that activate the stroma. Immune evasion via GM-CSF–driven myeloid recruitment and PD-L1 upregulation. Essentially every axis is downstream of KRAS, but growth drive is where the effect is strongest and most direct.

MYC (amplification / transcriptional upregulation)

GD	DR	CCB	MI	PSS	IET
↑↑↑	↑	↓↓↓	—	↑	↑

- **Confidence:** HIGH for GD, CCB. MED for DR, PSS, IET.
- **Caveat:** MYC is primarily a transcriptional amplifier of proliferation. It overrides cell-cycle checkpoints almost completely (hence ↓↓↓ on braking). Death resistance effect is paradoxical: MYC *sensitizes* to apoptosis in normal cells (oncogene-induced apoptosis), but in the context of

TP53 loss this brake is released, so the net effect in PDAC is mild pro-survival. Pro-stroma signaling is indirect via metabolic demand creating hypoxia. MYC does not directly drive motility.

EGFR (overexpression / autocrine activation)

GD	DR	CCB	MI	PSS	IET
↑↑	↑	↓	↑	—	—

- **Confidence:** MED across all axes.
- **Caveat:** EGFR is paradoxically important in PDAC biology but a clinical disappointment as a drug target (erlotinib has marginal benefit). Its growth drive effect is real but largely redundant with constitutive KRAS — it matters most as the *input receptor* that the EA can theoretically target. Motility effect is via EGFR→RAS→RAC pathway. In the Boolean network, EGFR functions primarily as a modulatable input node rather than a dominant driver.

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2 — DEATH RESISTANCE AXIS

TP53 (loss-of-function / dominant-negative missense)

GD	DR	CCB	MI	PSS	IET
↑	↑↑↑	↓↓↓	↑↑	↑	↑

- **Confidence:** HIGH for DR, CCB. MED for MI, PSS, IET.

- **Caveat:** TP53 loss does far more than block apoptosis. It demolishes G1/S and G2/M checkpoints (↓↓↓ braking), enables genomic instability, and in its gain-of-function missense forms (R175H, R273H — common in PDAC) *actively promotes* invasion via integrin recycling and RhoA signaling. The pro-stroma effect is indirect: TP53 loss cooperates with KRAS to amplify TGF-β and SHH secretion. **Critical modeling note:** TP53 gain-of-function mutations are qualitatively different from simple loss. For Cycle 1, treat as loss-of-function (simpler) but flag for future refinement.

BCL-XL / BCL2L1 (overexpression)

GD	DR	CCB	MI	PSS	IET
—	↑↑↑	—	—	—	—

- **Confidence:** HIGH for DR. HIGH for all others being negligible.
- **Caveat:** BCL-XL is a clean, single-axis gene. It sequesters BAX/BAK at the mitochondrial membrane, preventing cytochrome c release. No meaningful crosstalk to other axes. This is exactly why it's useful in the Boolean network: a dedicated survival node with a druggable handle (navitoclax/ABT-263, venetoclax-related compounds). **Important:** BCL-XL inhibition causes thrombocytopenia — the fitness function should penalize BCL-XL-targeting interventions with a toxicity cost.

BAX (typically not mutated; functionally suppressed by BCL-XL)

GD	DR	CCB	MI	PSS	IET
—	↓↓ (when suppressed)	—	—	—	—

- **Confidence:** HIGH.
- **Caveat:** BAX is rarely mutated in PDAC. Instead it's functionally neutralized by BCL-XL overexpression. In the Boolean network, BAX is the pro-death counterweight: apoptosis fires when BAX > BCL-XL. BAX is not a drug target itself — it's the node you *want* to be active. Modeling note: BAX should be ON by default, with its phenotypic effect gated by the BCL-XL ratio.

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3 — CELL CYCLE BRAKING AXIS

CDKN2A / p16 (homozygous deletion / promoter methylation)

GD	DR	CCB	MI	PSS	IET
↑↑	—	↓↓↓	—	—	—

- **Confidence:** HIGH.
- **Caveat:** CDKN2A loss is the earliest genetic event after KRAS in PDAC progression (PanIN-2 stage). Its effect is focused: it removes the p16→CDK4/6→RB brake. The growth drive increase is a *consequence* of losing the brake, not independent mitogenic signaling. This distinction matters for the Boolean network: CDKN2A=OFF should release the RB node, not directly activate MYC. No meaningful effect on stroma, invasion, or death.

SMAD4 / DPC4 (homozygous deletion / inactivating mutation)

GD	DR	CCB	MI	PSS	IET
↑	—	↓↓	↑↑	◆	↑

- **Confidence:** HIGH for CCB, MI, PSS. MED for IET.
- **Caveat — this is the most important entry in the entire map:** SMAD4 loss creates an *asymmetric* TGF-β response:
  - **In tumor cells:** TGF-β can no longer signal growth arrest (brakes broken) or apoptosis. But SMAD4-independent TGF-β signaling (via MAPK, RhoA) is intact, so TGF-β now *promotes* invasion and EMT in tumor cells.
  - **In stromal cells:** SMAD4 is intact. TGF-β fully activates PSCs → CAFs → ECM production.
  - **Net effect on PSS:** ◆ context-dependent. SMAD4 loss does NOT reduce TGF-β secretion — the tumor still pumps out TGF-β (driven by KRAS). But the tumor no longer *responds* to it as a brake. So the stroma-building loop continues unimpeded while the tumor escapes growth arrest. The pro-stroma signal doesn't change; the tumor's vulnerability to it disappears.

  Boolean network implication: SMAD4=OFF should disconnect TGF-β from the arrest/apoptosis outputs in tumor cells while leaving TGF-β→stroma activation fully intact. This is the single most critical asymmetry in the whole simulation.

RB1 (functional inactivation via CDKN2A loss / CDK4 hyperactivity)

GD	DR	CCB	MI	PSS	IET
↑	—	↓↓	—	—	—

- **Confidence:** HIGH.
- **Caveat:** RB1 is rarely deleted directly in PDAC — it's functionally inactivated by upstream loss of p16 (CDKN2A) or CDK4/6 hyperactivity. In the Boolean network, RB1 is the *convergence node*: multiple brake signals feed into it (p16, TGF-β/SMAD, DNA damage). RB1=OFF is the operational readout for "brakes are broken." Model it as a downstream integrator, not an independent input.

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4 — INVASION & MOTILITY AXIS

ZEB1 (transcriptional upregulation)

GD	DR	CCB	MI	PSS	IET
↓	↑	—	↑↑↑	—	↑↑

- **Confidence:** HIGH for MI. MED for DR, IET. LOW for GD.
- **Caveat:** ZEB1 is the EMT master switch. When ON, it represses CDH1 (E-cadherin), activates vimentin, and shifts the cell to a mesenchymal phenotype. The growth *decrease* reflects the well-documented "go vs. grow" tradeoff: EMT cells migrate more but proliferate less. Death resistance comes from EMT-associated anoikis resistance. Immune evasion via PD-L1 upregulation (ZEB1 directly binds the CD274 promoter). **Boolean network:** ZEB1 should be activated by TGF-β (SMAD-independent arm) and hypoxia (HIF1A), and its main output is CDH1=OFF.

CDH1 / E-cadherin (transcriptional repression by ZEB1, rarely mutated)

GD	DR	CCB	MI	PSS	IET
—	—	—	↑↑↑ (when lost)	—	—

- **Confidence:** HIGH.
- **Caveat:** CDH1 is an output node, not a driver. It's the *readout* of EMT status. CDH1=OFF means cell-cell adhesion is lost, contact inhibition is impaired, and motility is unrestricted. In PhysiCell terms: CDH1 state maps directly to `cell_cell_adhesion_strength`. CDH1 is not independently mutated in most PDAC — it's silenced by ZEB1. Model it as a downstream Boolean output, not an independent input.

MMP2 (transcriptional upregulation by stroma and tumor)

GD	DR	CCB	MI	PSS	IET
—	—	—	↑↑↑	↓	—

- **Confidence:** HIGH for MI. MED for PSS.
- **Caveat:** MMP2 degrades type IV collagen, laminin, and other ECM components. This is the *stroma-degrading* effector — the thing the EA might want to upregulate to break the barrier. The mild *decrease* in PSS reflects the fact that ECM degradation opposes stroma accumulation. However, MMP2 also releases ECM-sequestered TGF-β, which can paradoxically *increase* stroma activation. **Modeling decision:** For Cycle 1, treat MMP2 as net stroma-degrading (its mechanical ECM-clearing effect dominates). Flag the TGF-β liberation effect for Cycle 2 refinement.

ACTA2 / α-SMA (upregulation in activated stellate cells)

GD	DR	CCB	MI	PSS	IET
↑ (stroma)	↑ (stroma)	—	↑↑ (stroma)	↑↑↑	↑

- **Confidence:** HIGH for PSS, MI. MED for GD, DR, IET.
- **Caveat:** ACTA2 is the operational definition of CAF activation. This is a *stromal cell* gene. ACTA2=ON means the PSC has differentiated into a myofibroblastic CAF: it contracts ECM, produces collagen, secretes growth factors, and becomes resistant to apoptosis. The immune evasion effect is via CAF-mediated exclusion of T cells from the tumor nest. **Boolean network:** ACTA2 should be activated by TGF-β and SHH signaling in stromal cells only. Its outputs: COL1A1 production ↑, HAS2 production ↑, contractility ↑.

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5 — SECRETION & COMMUNICATION AXIS

TGFB1 (overexpression by tumor and stroma)

GD	DR	CCB	MI	PSS	IET
◆	—	◆	↑↑	↑↑↑	↑↑↑

- **Confidence:** HIGH across all axes.
- **Caveat — the dual-nature cytokine:**
  - In SMAD4-intact tumor cells: TGF-β = growth arrest + apoptosis (GD ↓, CCB ↑). Tumor suppressor.

- In SMAD4-null tumor cells (most late PDAC): TGF-β = invasion via SMAD-independent pathways (GD ──, CCB ──). The brake is gone.
- In stromal cells: TGF-β = full activation → ECM, contractility, more TGF-β. The barrier builder.
- Immune evasion: TGF-β is one of the most potent immunosuppressive cytokines (Treg induction, CD8 suppression).

Boolean network: TGFB1 is a *substrate*, not an intracellular node. Model it as a diffusible field. Its *effect* on any given cell depends on that cell's SMAD4 status. This is the core asymmetry.

SHH (overexpression by tumor)

GD	DR	CCB	MI	PSS	IET
── (tumor)	──	──	──	↑↑↑	↑

- **Confidence:** HIGH for PSS. MED for IET. HIGH for all others being negligible.
- **Caveat:** SHH in PDAC is almost purely paracrine: tumor secretes, stroma responds. The tumor itself has very low Hedgehog pathway activity. SHH drives CAF proliferation and ECM production via GLI1 in stromal cells. Immune evasion effect is indirect (stroma as physical barrier to immune infiltration). **Clinical trial warning:** Vismodegib (SHH inhibitor) failed catastrophically in PDAC trials — tumors actually *accelerated*. This may be because stroma also restrains the tumor. The fitness function must be able to detect this failure mode: stroma removal without tumor kill = bad outcome.

GLI1 (transcriptional activation in stroma, downstream of SHH)

GD	DR	CCB	MI	PSS	IET
↑↑ (stroma)	↑ (stroma)	—	—	↑↑	—

- **Confidence:** HIGH for GD (stroma), PSS. MED for DR.
- **Caveat:** GLI1 is the effector transcription factor of Hedgehog signaling in CAFs. It drives CAF proliferation (more fibroblasts = more barrier) and ECM gene expression. Including GLI1 separately from SHH allows the EA to test interventions at two levels of the same pathway (ligand vs. transcription factor), which is important because they have different pharmacological profiles (anti-SHH antibody vs. GANT61).

HAS2 (upregulation in activated CAFs)

GD	DR	CCB	MI	PSS	IET
—	—	—	—	↑↑↑	↑

- **Confidence:** HIGH for PSS. MED for IET.
- **Caveat:** HAS2 produces hyaluronic acid (HA), which does two things to the stroma: (1) increases interstitial fluid pressure (physically compresses vasculature, blocks drug delivery), and (2) creates a hydrated gel matrix that impedes diffusion. HA is the reason PDAC has such extraordinarily high interstitial pressure. Druggable via PEGPH20 (hyaluronidase), which showed initial promise but failed in Phase 3 (HALO-301) — likely because HA removal alone is insufficient without addressing collagen and immune evasion. **PhysiCell mapping:** HAS2 output should modify the diffusion coefficient of the drug substrate in the microenvironment field.

COL1A1 (upregulation in activated CAFs)

GD	DR	CCB	MI	PSS	IET
—	—	—	—	↑↑↑	↑

- **Confidence:** HIGH for PSS. MED for IET.
- **Caveat:** COL1A1 is the fibrillar collagen component of the stroma barrier. Distinct from HA (HAS2) in that collagen provides structural rigidity and alignment ("collagen highways" that can paradoxically guide tumor invasion along fiber tracks). Immune evasion effect is physical: dense collagen prevents T cell migration into tumor nests. **Modeling note:** COL1A1 and HAS2 together produce the ECM density field. Consider making them additive contributors to a single "ECM density" substrate rather than tracking them as separate fields — this reduces PDE complexity without losing the ability to target them independently via the EA.

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6 — STRESS ADAPTATION AXIS

HIF1A (stabilization under hypoxia)

GD	DR	CCB	MI	PSS	IET
◆	↑↑	—	↑↑	↑↑	↑↑

- **Confidence:** HIGH across all axes.

- **Caveat:** HIF1A is the hypoxia master regulator. Its effect on growth is context-dependent: acute hypoxia can slow proliferation, but HIF1A-adapted cells upregulate glycolysis and survive. In PDAC, HIF1A is frequently active because the stroma creates a hypoxic microenvironment — this creates a *feedback loop*: stroma → hypoxia → HIF1A → more stroma-building signals + therapy resistance. Death resistance via BCL-XL upregulation and autophagy. Motility via direct transcriptional activation of ZEB1. Pro-stroma via VEGF (not modeled in Cycle 1) and connective tissue growth factor. Immune evasion via adenosine pathway and PD-L1. **Boolean network:** HIF1A should be an input node gated by local O_2 concentration from the microenvironment field. O_2 below threshold → HIF1A=ON → multiple downstream effects.

NRF2 / NFE2L2 (gain-of-function mutation or transcriptional activation)

GD	DR	CCB	MI	PSS	IET
↑	↑↑	—	—	—	—

- **Confidence:** MED for GD, DR. HIGH for others being negligible.
- **Caveat:** NRF2 is the xenobiotic/oxidative stress response master regulator. In PDAC, NRF2 activation (sometimes via KRAS-driven transcription, sometimes via KEAP1 mutation) confers resistance to gemcitabine and other chemotherapeutics. The mechanism is specific: NRF2 upregulates drug metabolizing enzymes and efflux pumps (→ ABCB1). Mild growth promotion via metabolic reprogramming (pentose phosphate pathway, glutathione synthesis). **Modeling note:** NRF2 should activate ABCB1 as its primary simulation-relevant output. The growth effect is secondary and can be omitted if Boolean network complexity needs trimming.

ABCB1 / MDR1 (transcriptional upregulation)

GD	DR	CCB	MI	PSS	IET
—	↑↑↑ (drug-specific)	—	—	—	—

- **Confidence:** HIGH for DR. HIGH for others being negligible.
- **Caveat:** ABCB1 is a drug efflux pump — it physically pumps chemotherapy out of the cell. Its "death resistance" effect is entirely drug-specific: it reduces intracellular drug concentration. It provides zero survival advantage in the absence of drug. **Critical modeling note:** ABCB1's effect should be conditional on drug presence. Boolean logic: ABCB1=ON AND drug_present → intracellular_drug_concentration ↓↓. This makes it an *induced* resistance mechanism, not a constitutive one. In PhysiCell, map this to a modifier on the drug uptake/effect calculation, not to the basal apoptosis rate.

Master Summary Table

#	Gene	Mutation	GD	DR	CCB	MI	PSS	IET	Conf.
1	KRAS	GoF	↑↑↑	↑↑	↓↓	↑	↑↑↑	↑↑	HIGH
2	MYC	Amp/up	↑↑↑	↑	↓↓↓	—	↑	↑	HIGH/MED
3	EGFR	Overexpr	↑↑	↑	↓	↑	—	—	MED
4	TP53	LoF	↑	↑↑↑	↓↓↓	↑↑	↑	↑	HIGH
5	BCL-XL	Overexpr	—	↑↑↑	—	—	—	—	HIGH
6	BAX	Suppressed	—	↓↓*	—	—	—	—	HIGH

#	Gene	Mutation	GD	DR	CCB	MI	PSS	IET	Conf.
7	CDKN2A	HomDel	↑↑	—	↓↓↓	—	—	—	HIGH
8	SMAD4	LoF	↑	—	↓↓	↑↑	◆	↑	HIGH
9	RB1	Func.inact	↑	—	↓↓	—	—	—	HIGH
10	ZEB1	Up	↓	↑	—	↑↑↑	—	↑↑	HIGH/MED
11	CDH1	Repressed	—	—	—	↑↑↑	—	—	HIGH
12	MMP2	Up	—	—	—	↑↑↑	↓	—	HIGH/MED
13	ACTA2	Up [S]	↑	↑	—	↑↑	↑↑↑	↑	HIGH
14	TGFB1	Overexpr	◆	—	◆	↑↑	↑↑↑	↑↑↑	HIGH
15	SHH	Overexpr	—	—	—	—	↑↑↑	↑	HIGH
16	GLI1	Active [S]	↑↑	↑	—	—	↑↑	—	HIGH/MED
17	HAS2	Up [S]	—	—	—	—	↑↑↑	↑	HIGH
18	COL1A1	Up [S]	—	—	—	—	↑↑↑	↑	HIGH
19	HIF1A	Stabilized	◆	↑↑	—	↑↑	↑↑	↑↑	HIGH
20	NRF2	GoF/up	↑	↑↑	—	—	—	—	MED
21	ABCB1	Up	—	↑↑↑†	—	—	—	—	HIGH

> * BAX row shows the effect of its *suppression* (the PDAC-relevant state), not its activation. † ABCB1 death resistance is conditional on drug presence; no survival benefit at baseline. ◆ = context-dependent; see individual gene entry for explanation. [S] = effect manifests in stromal cells, not tumor.

Critical Caveats for Implementation

1. The SMAD4 Asymmetry is Non-Negotiable

TGF-β must have *opposite* effects depending on SMAD4 status. If the Boolean network treats TGF-β as a uniform signal, the stroma barrier will not emerge correctly. This is the single most important biological feature of the entire simulation.

2. Context-Dependent (◆) Entries Require Conditional Logic

KRAS-driven TGF-β secretion is unconditional → the tumor *always* builds its own prison. But TGF-β's *effect* on the tumor depends on SMAD4. TGFB1 and HIF1A rows cannot be reduced to static arrows — they require if/then logic in the Boolean network.

3. The Go-vs-Grow Tradeoff (ZEB1/MYC Tension)

EMT (ZEB1=ON) and maximal proliferation (MYC high) are partially antagonistic. Cells shouldn't simultaneously be maximally migratory AND maximally proliferative. The Boolean network should enforce this: ZEB1=ON should apply a mild penalty to cycle transition rate.

4. Drug-Conditional Effects (ABCB1)

ABCB1 does nothing at baseline. Its death resistance effect activates only when the drug substrate is present above a threshold. This prevents the EA from seeing ABCB1 as a constitutive survival gene and ensures the simulation correctly captures *induced* resistance.

5. Stroma Genes Are Stromal-Cell-Only

ACTA2, GLI1, HAS2, COL1A1 should only be active in the stromal cell Boolean network. If these accidentally leak into tumor cell logic, the simulation will produce tumor cells that build their own stroma — biologically wrong and will confuse the EA.

6. Clinical Trial Failures as Validation Benchmarks

Three anti-stroma therapies have failed in PDAC:

- **Vismodegib** (SHH inhibitor): tumors accelerated. Model must reproduce this.
- **PEGPH20** (hyaluronidase): insufficient alone. Model should show partial barrier removal.
- **Nab-paclitaxel** (stroma-depleting chemo): works only in combination. Model should show synergy requirement.

If the simulation doesn't reproduce these known failure modes, the gene→behavior mapping has a bug.

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21 genes. 6 axes. No numbers — just arrows and logic. The numbers come when we parameterize the PhysiCell phenotype mappings.